

Introduction:

NM0221-I is a SOC-based low-power wireless NIC Card by Nebulae that provides long-range connectivity for IoT devices. This NIC Card is engineered to support 6LoWPAN and IEEE 802.15.4 PHY network. NM0221-I is ideal for applications which need self-forming and self-healing mesh networks. The NIC Card is ideal for IS16444 smart metering applications because it is equipped with sufficient memory and is WPC certified. NM0221-I's FOTA capability makes it suitable for remote security updates. By leveraging 6LoWPAN technology, it enables seamless communication between meters and utility providers, facilitating advanced smart grid capabilities. With features such as mesh networking for reliable connectivity, long-range communication, low power consumption, and real-time monitoring, the NIC card offers valuable insights into energy usage patterns. It provides a comprehensive solution for modernizing metering infrastructure, driving efficiency, reliability, and sustainability in the energy sector.

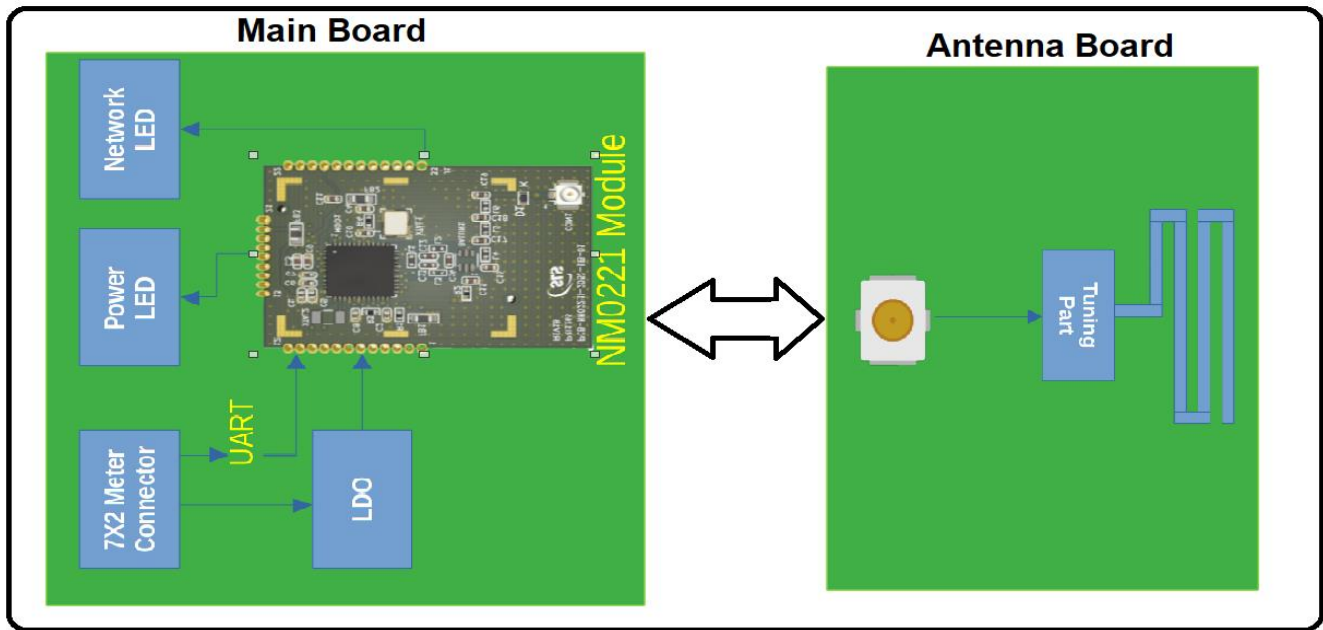
Features:

- NM0221-I WPC certified Based NIC Card
- Frequency band: 863 MHz to 876 MHz
- Up to 1024 kB Flash Program Memory
- Up to 128 kB RAM Data Memory
- Support BPSK / DBPSK TX modulation format
- 2 × Watchdog timer with dedicated RC oscillator
- IEEE 802.15.4 compliant
- Two LED is used one is for Power and another is for Network
- Easy to install.
- Provides an OTA (Over the Air) firmware update mechanism

Technical Specification:

- Operating Voltage 3.3V to 5V DC Input Supply
- Working temperature 0~70 Celsius (commercial grade) or -40~85 Celsius (industrial grade)
- Dimensions 62.8 X 42.8 mm
- 2-layer PCB design
- Radiated Transmit Power 15dBm
- RX Sensitivity (@ 1% PER) 38.4 kbps -105 dBm / 100 kbps -101 dBm
- 2 Wire serial UART Communication for Meter interface
- Current consumption 100mA@3.3V TX mode

Functional Block Diagram:



Meter Pin Connection

Sr No.	Meter Pin Number	Meter Pin Name	NM0221-I module Pin number
1	1,2,4,5,8,9,11	NC	-
2	3	Power EN(Optional with NL)	1
3	6	Meter TX	26
4	7	Meter RX	8
5	10	NIC Reset(Optional With NL)	27
6	13,14	GND	-

Figure 1: IO Details